Second-Order Balanced Truncation for Passive Order Reduction of RLC Circuits

Boyuan Yan and Sheldon X.-D. Tan
University of California,
Riverside, CA, 92521

ABSTRACT

This paper presents a new second-order balanced passive order reduction method for RLC and RCS circuits. The new method, called SBPOR, combines both second order balanced truncatoin with congruence transformation to produce passive compact models. This is enabled by the exploitation of the spectrial structure of RCS circuit matrices in MNA form. The new method has the same accuacry as the standard second order truncation balanced realization (TBR) methods given the same order. But it can preserve the passivity and structure related properties like reciporcity. Also SBPOR is computationally more efficient than standard TBR methods as it only need to solve one linear matrix equation in contrast to solving two linear matrix equations for general TBR methods and solving two nonlinear matrix equartions for passive TBR methods. SBPOR is the first passive second order TBR method proposed so far. Experimental results show that SBPOR is more accurate than stable second order TBR method and is superior to the latest projection based second order reduction algorithm SAPOR.

Keywords

model order reduction, second-order balanced truncation, passivity, reciprocity

1. INTRODUCTION

Model order reduction (MOR) is an efficient technique to reduce the circuit complexity while producing a good approximation of the input and output behavior. From formulation point of view, MOR techniques can be classified into first-order based methods (using modified nodal analysis, MNA) and second-order based methods (using nodal analysis, NA). In terms of projection subspace, these approaches are divided into two broad categories, namely moment-matching based methods and balanced truncation based methods. In the former case, the system is projected onto a subspace to match dominant moments while in the latter case system is projected onto a subspace both easily controllable and easily observable.

Moment-matching based approaches have been a great success in this field in the past decade. Those methods perform implicit moment matching in a projection framework and stability, passivity and structure information inherent to RLC circuits can be preserved easily by exploiting the special properties of RLCS circuit matrices. As for moment-matching based MOR in first-order formulation, AWE [8] was proposed first, which suffers from numerical instability due to explicit moment-matching. To solve this problem, Krylov-subspace based methods [1, 10] were proposed, where implicit moment-matching is carried out in a projection framework. Furthermore, to ensure the stability of the simulation process, PRIMA [6] was developed based on Arnoldi process. PRIMA exploits the positive semi-definiteness property of RLC matrices in MNA formulation and it can ensure the passivity of the reduced models via a congruency transformation. More recently, SPRIM [2] further exploit the block structure of RLC formulation such that, in addition to passivity, structure information like symmetry inherent to RLC circuits can be preserved at the same time.

Recently, NA based second-order MOR has been proved to be more suitable for RCS reduction. RCS circuits are more suitable for modeling the magnetic effects as susceptance is shown to be ambient to the reduction than inductance. The reason is that the susceptance matrix in NA is diagonally dominant and can be sparsified by a simple truncation method without disrupting the positive definiteness. Another benefit by using NA is that circuit matrices for RCS are symmetric and positive-definite LU decomposition can be performed more efficiently. In the past several years, projection based second order reduction methods have been proposed ranging from explicit moment matching approach ENOR [9] to implicit moment-matching approach SAPOR [12].

But to obtain wideband accurate models, which is specailly true for inductively coupling circuits, TBR based reducton approaches, which have global error bounds, are better solutions. Truncated balanced realization (TBR) methods, which has well-developed in the control community [5], has been studied recently in VLSI design automation community and passive TBR methods have also been proposed [7, 13]. So far, existing TBR-like methods for RLC reduction are all first-order based and working on a standard state-space model without exploiting any additional information about the state matrices. The passive TBR methods are more expensive as they need to solve two quadratic matrix equations. Also all the structure infromattion inherent to RCS circuit matrices such as symmtric, positive semi-definiteness will be lost during the reduction process.

Recently Meyer and Srinivasan [4] proposed to combine second order TBR with congruence transformation to perserve the symmtry and thus stability of the reduced systems. But this method can lead to accucary loss compared with standard second order TBR methods. The reason is that congruence transformation used does not equal to similarity transformation used in standard TBR method, which will lead to unbalanced transformed systems.

In this paper, we develop an novel passive balanced truncation method for RCS reduction, SBPOR, which is based on a second-order form. The new approach overcome the problems in [4] by defining second-order gramians based on a symmetric first-order realization in descriptor form for RCS circuits formulated in NA form. Our main contribuations are as follows: (1) we show for the first time for a general RLC/RCS circuit, we can perform passive balanced truncation reduction without using positive real balanced truncation, which is more expensive. The new algorithm can achieve a real balance for the transformed system, which means *no accuacry loss* compared to standard second order TBR method. (2) The new algorithm preserves symmetry, passivity, and reciprocity simultaneously, which is not possible with existing passive TBR methods. (3) The SBPOR only needs to solve only one Lyapunov-like equation, which is in contrast to solving two nonlinear matrix equations in existing passive TBR methods. Experimental results show that the proposed method is much accurate than the projection-based second order reduction method SAPOR over wide frequency ranges.

This paper is organized as the following: In Section 2, we review the standard TBR method and its passive version. In Section 3, we introduce the existing second-order balanced truncation approach and point out some drawbacks of this method. Our new approach, SBPOR, is described in Section 4. Several experimental results are reported in Section 5 to demonstrate the effectiveness of our proposed method. Section 6 concludes the paper.

2. FIRST-ORDER BALANCED TRUNCATION

In this section, we briefly review standard TBR and its passive version in a first-order form, and then point out why they are not suitable for RLC reduction.

2.1 Standard balanced truncation

Given a system in a standard state-space form (A,B,C,D)

$$\begin{aligned} \dot{x}(t) &= Ax(t)+Bu(t) \\ y(t) &= Cx(t)+Du(t) \end{aligned} \tag{1}$$

where $A \in R^{n\times n}$, $B \in R^{n\times p}$, $C \in R^{p\times n}$, $D \in R^{p\times p}$, $y(t)$, $u(t) \in R^p$. The controllable and observable gramians are the unique symmetric positive definite solutions to the Lyapunov equations.

$$\begin{aligned} AW_c+W_cA^T+BB^T=0 \\ A^TW_o+W_oA+C^TC=0 \end{aligned} \tag{2}$$

Since the eigenvalues of the product W_cW_o are invariant under similarity transformation, we can perform a similarity transformation $(A_b=T^{-1}AT, B_b=T^{-1}B, C_b=CT)$ to diagonalize the product W_cW_o

$$\tilde{W_c}\tilde{W_o}=T^{-1}W_cW_oT=diag(\sigma_1{}^2,\sigma_2{}^2,\ldots,\sigma_n{}^2) \tag{3}$$

The Hankel singular values of the system, σ, are the square roots of the eigenvalues of the product W_cW_o. After the transformation, $\tilde{W}_c$ and $\tilde{W}_o$ are equal and diagonal and such a state-space form (A_b,B_b,C_b,D) is called balanced

$$W_{bc}=W_{bo}=\Sigma=diag(\sigma_1,\sigma_2,\ldots,\sigma_n) \tag{4}$$

The Hankel singular values as well as the eigenvalues of gramian product, characterize the 'importance' of state variables. States of the balanced system corresponding to the small Hankel singular values are difficult to reach and to observe at the same time. Such states are less involved in the energy transfer from inputs to outputs. Therefore, a general idea of balanced truncation is to transform the system into a balanced form (A_b,B_b,C_b,D) and to truncate the states that correspond to the small Hankel singular values. We may partition Σ into

$$\Sigma=\begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} \tag{5}$$

Conformally partition the transformed matrices as

$$A_b=\begin{bmatrix} A_{b11} & A_{b12} \\ A_{b21} & A_{b22} \end{bmatrix} \quad B_b=\begin{bmatrix} B_{b1} \\ B_{b2} \end{bmatrix} \quad C_b=\begin{bmatrix} C_{b1} & C_{b2} \end{bmatrix} \tag{6}$$

The reduced model of order r is obtained by taking the $r\times r$, $r\times p$, $p\times r$ leading blocks of A_b, B_b, C_b, respectively. This truncation leads to a balanced reduced-order system $(A_{b11},B_{b1},C_{b1},D)$.

2.2 Positive-real balanced truncation

Standard TBR method does not ensure the passivity of the reduced models. To mitigate this problem, a positive-real TBR (PR-TBR) [7] was proposed to generate the passive models by solving the Lur'e equations

$$\begin{aligned} AX_c+X_cA^T &= -K_cK_c^T \\ X_cC^T-B &= -K_cJ_c^T \\ J_cJ_c^T &= D+D^T \end{aligned} \tag{7}$$

and

$$\begin{aligned} A^TX_o+X_oA &= -K_o^TK_o \\ X_oB-C^T &= -K_o^TJ_o \\ J_o^TJ_o &= D+D^T \end{aligned} \tag{8}$$

where the solutions X_c and X_o are analogous to the controllability gramian and observability gramian respectively. Note that, Lur'e equations are quadratic matrix equations and thus more expensive than linear matrix equations like Lyapunov equations.

2.3 Drawbacks of first-order TBR in RLC reduction

Consider the RLC circuit in MNA formulation, which is in a first-order form

$$\begin{aligned} &C\dot{x}(t)=-Gx(t)+Bu(t) \\ &y(t)=B^Tx(t) \end{aligned} \tag{9}$$

where G and C are positive-semidefinite, sparse, and have special block structure. In fact, such a form is provably passive. Moment-matching based approaches [2] take advantage of the properties so that structure as well as passivity can be preserved easily in the reduction process. However, when applying existing TBR methods [7, 13], the first thing to do is to convert MNA formulation into a standard state-space model(1)

$$\begin{aligned} &\dot{x}(t)=-C^{-1}Gx(t)+C^{-1}Bu(t) \\ &y(t)=B^Tx(t) \end{aligned} \tag{10}$$

All the nice properties available are lost in this process. Also, this form is computationally undesirable since $C^{-1}G$ may be dense even though C and G are sparse. Moreover, $C^{-1}G$ will be computationally ill-determined if C is nearly singular. In addition, the passivity is preserved by solving two expensive quadratic matrix equations and the structure information inherent to RLC circuits cannot be preserved in the reduction process.

3. SECOND-ORDER BALANCED TRUNCATION IN THE CONTROL AREA

In this section, we will review second-order balanced truncation approach in pioneering work [4] and point out some drawbacks of this method.

3.1 Second-order model order reduction

Given a system in a standard second-order form

$$\begin{aligned} &M\ddot{q}(t)+D\dot{q}(t)+Kq=B_2u(t) \\ &y(t)=Pq(t)+Q\dot{q}(t) \end{aligned} \tag{11}$$

where $u(t)\in R^m$, $y(t)\in R^p$, and the coordinates $q(t)\in R^n$. For most physical systems, the following conditions hold

$$M=M^T>0 \quad K=K^T\geq 0 \quad D=D^T\geq 0 \tag{12}$$

which means M, D, and K are symmetric positive semi-definite. The problem of interest is to obtain a reduced order model in second-order form

$$\begin{aligned} &M_r\ddot{q}_r(t)+D_r\dot{q}_r(t)+K_rq_r(t)=B_{2r}u(t) \\ &y(t)=P_rq_r(t)+Q_r\dot{q}_r(t) \end{aligned} \tag{13}$$

where $q_r(t)\in R^r$, $r<n$. We want the input/output (I/O) behavior of the reduced model to be close to the I/O behavior of the original model. In addition, we want the conditions (12) to hold for the reduced model.

3.2 Second-order gramians and balancing

In [4], second-order gramians are defined based on a first-order realization with 2n-dimensional state $x^T=[q^T\,\dot{q}^T]$

$$\begin{aligned} \dot{x}(t) &= Ax(t)+Bu(t) \\ y(t) &= Cx(t) \end{aligned} \tag{14}$$

where

$$A = \begin{bmatrix} 0 & I \\ -M^{-1}K & -M^{-1}D \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ M^{-1}B_2 \end{bmatrix} \quad C = \begin{bmatrix} P & Q \end{bmatrix} \tag{15}$$

The first-order realization has the same input-output behavior as the second-order system. Although a first-order MOR approach can be applied to reduce the realization, the reduced model is also in a first-order form. In order to directly define gramians for second-order systems, we recall the definition of gramians for first-order systems: the controllability gramian W_c arises from the optimization problem

$$\begin{array}{l} min_{u\in L_2[-\infty,0]} \left(J = \int_{-\infty}^{0} u^2(t)dt\right) \\ \text{subject to} \\ \dot{x}(t) = Ax(t) + Bu(t) \\ x(0) = x_0 \end{array} \tag{16}$$

and the optimum for this problem is

$$x_0^T W_c^{-1} x_0 \tag{17}$$

which is the minimum energy for the system to reach a specific state x_0 over all past inputs. Similarly, one of the analogous optimization problems associated with the second-order form can be formed as follows [4]

$$\begin{array}{l} min_{\dot{q}(0)\in R^n, u\in L_2[-\infty,0]} \left(J = \int_{-\infty}^{0} u^2(t)dt\right) \\ \text{subject to} \\ M\ddot{q}(t) + D\dot{q}(t) + Kq = B_2u(t) \\ q(0) = q_0 \end{array} \tag{18}$$

which minimizes the necessary energy to reach the given q_0 over all past inputs and initial $\dot{q}$. If we compatibly partition the controllability gramian of the first-order realization (14) as

$$W_c = \begin{bmatrix} R & S \\ S^T & T \end{bmatrix} \tag{19}$$

then the optimum for the problem (18) is

$$q_0^T R^{-1} q_0 \tag{20}$$

and thus the controllability gramian of the second-order system is

$$W_{2c} = R \tag{21}$$

Similarly, if we compatibly partition the observability gramian of the first-order realization (14) as

$$W_o = \begin{bmatrix} U & V \\ V^T & W \end{bmatrix} \tag{22}$$

then the observability gramian of the second-order system is

$$W_{2o} = U \tag{23}$$

The second-order gramians are symmetric positive definite and the eigenvalues of the gramian product $W_{2c}W_{2o}$ are invariant under a similarity transformation

$$\tilde{W}_{2c}\tilde{W}_{2o} = T^{-1}W_{2c}W_{2o}T = diag({\sigma_{21}}^2, {\sigma_{22}}^2, \ldots, {\sigma_{2n}}^2) \tag{24}$$

Similar to the first-order case, the second-order system can be balanced as

$$\begin{array}{l} M_b\ddot{q_b}(t) + D_b\dot{q_b}(t) + K_bq_b = B_{2b}u(t) \\ y(t) = P_bq_b(t) + Q_b\dot{q_b}(t) \end{array} \tag{25}$$

where

$$\begin{array}{c} M_b = T^{-1}MT, D_b = T^{-1}DT, K_b = T^{-1}KT \\ B_{2b} = T^{-1}B_2, P_b = PT, Q_b = QT \end{array} \tag{26}$$

However, in order to preserve the symmetry and thus stability of the original system, the truncation in [4] is performed on the following coordinate system instead, where

$$\begin{array}{c} M_b = T^TMT, D_b = T^TDT, K_b = T^TKT \\ B_{2b} = T^TB_2, P_b = PT, Q_b = QT \end{array} \tag{27}$$

Unfortuneately, the second-order gramians in this coordinate system are not diagonalized and the system is not really balanced. As a result, there is no telling what states are important in energy sense and the accuracy is sacrificed.

4. SBPOR

In this section, we propose our novel technique SBPOR to generate accurate reduced model with guaranteed stability, passivity, and reciprocity for RLC circuits.

4.1 Symmetric realization in descriptor form

As mentioned before, instead of MNA formulation, RLC circuits can be formulated as NA formulation, which is in a second-order form (11) with $M = C$, $D = G$, $K = \Gamma$, $P = 0$, $Q = B_2^T$

$$\begin{array}{l} C\ddot{q}(t) + G\dot{q}(t) + \Gamma q(t) = B_2u(t) \\ y(t) = B_2^T\dot{q}(t) \end{array} \tag{28}$$

where $u(t)$, $y(t) \in R^m$ are input currents and output voltages; $\dot{q(t)}$ are nodal voltages; G, C, Γ are matrices of conductance, capacitance and susceptance and

$$C = C^T \geq 0 \quad G = G^T \geq 0 \quad \Gamma = \Gamma^T \geq 0 \tag{29}$$

In this paper, instead of using the first-order realization (14), we choose another first-order realization in descriptor form [3] with 2n-dimensional state $x^T = [q^T \, \dot{q}^T]$

$$\begin{array}{l} E_d\dot{x}(t) = A_dx(t) + B_du(t) \\ y(t) = C_dx(t) \end{array} \tag{30}$$

where

$$\begin{array}{ll} E_d = \begin{bmatrix} -\Gamma & 0 \\ 0 & C \end{bmatrix} & A_d = \begin{bmatrix} 0 & -\Gamma \\ -\Gamma & -G \end{bmatrix} \\ B_d = \begin{bmatrix} 0 \\ B \end{bmatrix} & C_d = \begin{bmatrix} 0 & B^T \end{bmatrix} \end{array} \tag{31}$$

Since C, G, Γ are all symmetric, we have

$$E_d = E_d^T, A_d = A_d^T, B_d = C_d^T \tag{32}$$

4.2 Second-order balanced truncation

Controllability and observability gramians in descriptor form can be obtained from solving the following generalized Lyapunov equations [11].

$$\begin{array}{l} E_dW_{dc}A_d^T + A_dW_{dc}E_d^T + B_dB_d^T = 0 \\ E_d^TW_{do}A_d + A_d^TW_{do}E_d + C_d^TC_d = 0 \end{array} \tag{33}$$

Since $A_d = A_d^T$, $E_d = E_d^T$, $C_d = B_d^T$, it is easy to show that both gramians are the same. If we compatibly partition the gramians as

$$W_{dc} = W_{do} = \begin{bmatrix} R & S \\ S^T & T \end{bmatrix} \tag{34}$$

then the generalized second-order gramians arising from the optimization problem are

$$W_{d2c} = W_{d2o} = R \tag{35}$$

The gramians are symmetric positive definite and the eigenvalues of the product are I/O invariants independent of coordinate choice of the second-order system (28). In addition, since $W_{d2c} = W_{d2o}$, the gramian product is symmetric

$$(W_{d2c}W_{d2o})^T = W_{d2c}W_{d2o} \tag{36}$$

which means the product is orthogonally diagonalizable, i.e. there exists an orthogonal matrix $T(T^{-1} = T^T)$ such that

$$\tilde{W}_{d2c}\tilde{W}_{d2o} = T^T W_{d2c} W_{d2o} T = diag(\sigma_{d21}{}^2, \sigma_{d22}{}^2, \ldots, \sigma_{d2n}{}^2) \quad (37)$$

So the system can be balanced as

$$\begin{aligned} &C_b \ddot{q}_b(t) + G_b \dot{q}_b(t) + \Gamma_b q_b(t) = B_{2b} u(t) \\ &y = B_{2b}^T \dot{q}_b(t) \end{aligned} \quad (38)$$

where

$$C_b = T^T C T \quad G_b = T^T G T \quad \Gamma_b = T^T \Gamma T \quad B_{2b} = T^T B_2 \quad (39)$$

This kind of transformation is known as congruency transformation, which preserves symmetry and definiteness of matrices such that

$$C_b = C_b^T \geq 0 \quad G_b = G_b^T \geq 0 \quad \Gamma_b = \Gamma_b^T \geq 0 \quad (40)$$

In the balanced coordinate, we have

$$\tilde{W}_{d2c} = \tilde{W}_{d2o} = \Sigma = diag(\sigma_{d21}, \sigma_{d22}, \ldots, \sigma_{d2n}) \quad (41)$$

We may select the reduced order r and partition Σ into

$$\Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} \quad (42)$$

Conformally partition the transformed matrices as

$$\begin{aligned} C_b &= \begin{bmatrix} C_r & C_{b12} \\ C_{b21} & C_{b22} \end{bmatrix} \quad G_b = \begin{bmatrix} G_r & G_{b12} \\ G_{b21} & G_{b22} \end{bmatrix} \\ \Gamma_b &= \begin{bmatrix} \Gamma_r & \Gamma_{b12} \\ \Gamma_{b21} & \Gamma_{b22} \end{bmatrix} \quad B_b = \begin{bmatrix} B_r \\ B_{b2} \end{bmatrix} \end{aligned} \quad (43)$$

After truncation, a balanced reduced-order system is given as

$$\begin{aligned} &C_r \ddot{q}_r(t) + G_r \dot{q}_r(t) + \Gamma_r q_r(t) = B_{2r} u(t) \\ &y = B_{2r}^T \dot{q}_r(t) \end{aligned} \quad (44)$$

where

$$C_r = C_r^T \geq 0 \quad G_r = G_r^T \geq 0 \quad \Gamma_r = \Gamma_r^T \geq 0 \quad (45)$$

which means the reduced-order system has guaranteed stability, passivity, and reciprocity [9]. Compared with existing techniques, SBPOR shows for the first time that a second-order system for RLC circuits can be really balanced via congruency transformation. In addition, only one Lyapunov-like equation is needed to be solved. The basic algorithm flow for SBPOR is given in Fig. 1. Since TBR-like methods are expensive, we also propose the combined SBPOR and SAPOR flow in Fig. 2 to deal with large scale circuits.

ALGORITHM 1: SBPOR

Input: C, G, Γ, B Output: C_r, G_r, Γ_r, B_r

1. Form the symmetric first order realization in descriptor form(30)
2. Solve $E_d W_d^T A_d^T + A_d W_d E_d^T + B_d B_d^T = 0$ for W_d
3. Partition W_d as:
   $W_d = \begin{bmatrix} R & S \\ S^T & T \end{bmatrix}$
4. Compute Cholesky factors:
   $W_{d2c} = R = LL^T$
5. Compute SVD of Cholesky product:
   $U\Sigma V = L^T L$
6. Compute the balancing transformation
   $T = LV\Sigma^{-1/2}$
7. Form the balanced realization as
   $C_b = T^T CT, G_b = T^T GT, \Gamma_b = T^T \Gamma T, B_{2b} = T^T B_2$
8. Select reduced order r and partition C_b, G_b, Γ_b, B_b conformally
9. Truncate C_b, G_b, Γ_b, B_b to form the reduced realization C_r, G_r, Γ_r, B_r

Figure 1: The algorithm for SBPOR.

ALGORITHM 2: COMBINED SAPOR AND SBPOR

1. Perform SAPOR to obtain a small-size passivity-preserved initial reduced second-order model from a large-size original model.
2. Perform SBPOR to obtain a optimal passivity-preserved final reduced second-order model from the initial model.

Figure 2: Combined SAPOR and SBPOR methods.

5. EXPERIMENTAL RESULTS

In this section, we show examples that illustrate the effectiveness of proposed SBPOR method and compare it with existing MOR approaches. All the algorithms are implemented in Matlab 7.0

5.1 Comparison with first-order TBR

Given a circuit in the form (28), we compare the reduced first-order model of dimension $2q$ computed by the standard TBR applied to the equivalent first-order realization (14) and the reduced second-order model of dimension q computed by SBPOR applied to (28) directly. We choose a small circuit as an example so that both impedance and real part can be compared at all possible reduced orders. The RLC circuit has 4 nodal voltages and thus has a dimension of 4 in a second-order formulation. The equivalent first-order realization has a dimension of 8. As shown in Fig. 3-5, SBPOR outperforms standard TBR at each reduced order. This can be explained from the ”energy” distribution of singular values as shown in Fig. 6, where the second-order singular values decay much faster than the first-order ones. The passivity of reduced models can be tested from the real parts. As expected, SBPOR can guarantee the passivity of reduced models while standard TBR cannot. As shown in Fig. 3-5, only in Fig. 5, the real part of TBR reduced model is positive at all frequencies and thus the reduced model is passive while the others are not. Note that standard TBR applied to the equivalent first-order realization(14) also results in a first-order reduced model and thus is not a second-order MOR approach available. Since standard TBR in a first-order form is classic and usually more accurate than PR-TBR [7], we just use it as a criterion to show the accuracy of our new approach.

5.2 Comparison with SAPOR

In the second example, we want to compare our method with moment-matching based second-order MOR approach SAPOR [12]. The example is a RLC circuit, which has 100 nodal voltages. The reduced second-order model has a dimension of 2. As shown in Fig. 7, SBPOR is globally accurate at all frequencies while SAPOR has very good local behavior around DC but behaves so bad at other frequencies. The error is shown in Fig. 8, where the maximum absolute error for SBPOR is about 10 but for SAPOR is almost 100.

5.3 Comparison with existing second-order TBR

In this part, we want to compare the new method, SBPOR, with existing technique [4] in the control literature, which we name TBR2.

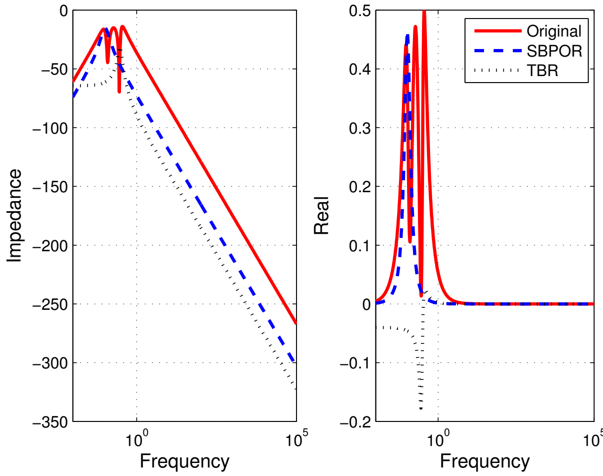

Figure 3: Comparison between 1 dimension SBPOR and 2 dimension TBR.

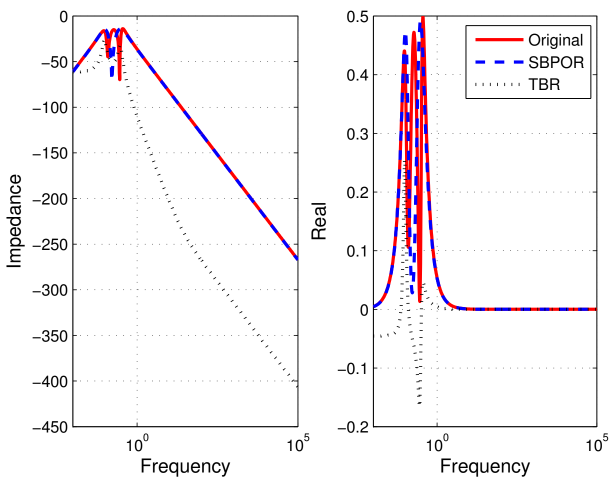

Figure 4: Comparison between 2 dimension SBPOR and 4 dimension TBR.

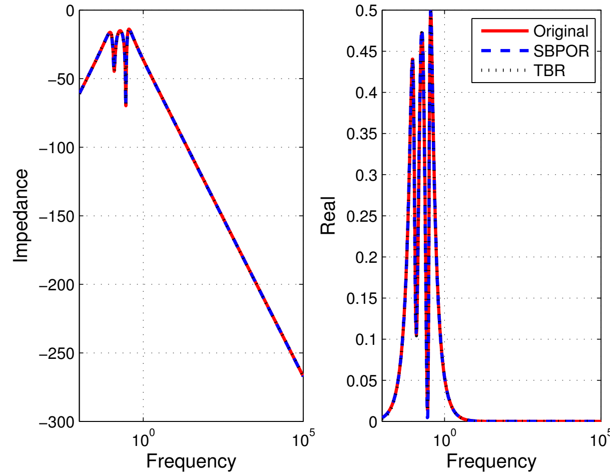

Figure 5: Comparison between 3 dimension SBPOR and 6 dimension TBR.

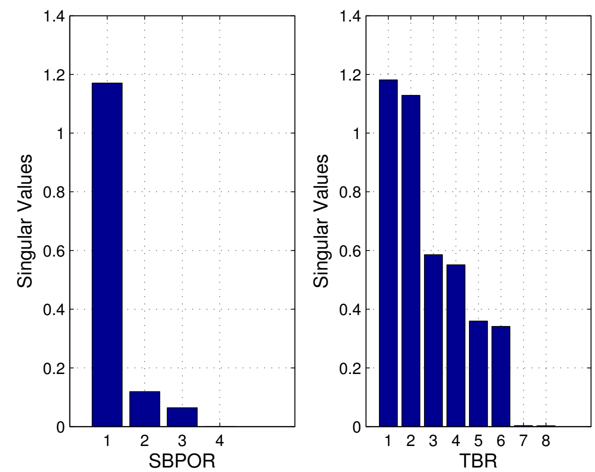

Figure 6: Comparison of singular values.

The example is a RLC circuit with 100 nodal voltages and the reduced dimension is 10. In Fig. 9, we can see that SBPOR outperforms TBR2 obviously. As shown in Fig. 10, the maximum absolute error for SBPOR is smaller than 10 while it is almost 100 for TBR2. The reason is that the system in TBR2 is not really balanced and thus the accuracy is sacrificed.

6. CONCLUSION

In this paper, we proposed a novel second-order balanced truncation for passive order reduction of RCS circuits, namely *SBPOR*. SBPOR combine the balanced truncation with congruence transformation to produce the reduced models. Different from the existing second order balanced truncation methods, the new approach can produce the passive reduced second order models by balanced truncation for the first time. It exploits the symmtric and positive semidefinite properties of the RCS circuit matrices in second form and symmtry-perserving first order realization in descriptor form to ensure the both passivity and balance are achieved by congruence transformation. SBPOR only requires solving one linear matrix equations instead of two nonlinear equations in existing passive TBR methods. Experimental results show that SBPOR is more accurate than stable second order TBR method and is more accurate than the best projection based second order reduction method SAPOR for the same sizes of reduced models.

7. REFERENCES

[1] P. Feldmann and R. W. Freund, “Efficient linear circuit analysis by pade approximation via the lanczos process,” *IEEE Trans. on Computer-Aided Design of Integrated Circuits and Systems*, vol. 14, no. 5, pp. 639–649, May 1995.

[2] R. W. Freund, “SPRIM: structure-preserving reduced-order interconnect macromodeling,” in *Proc. Int. Conf. on Computer Aided Design (ICCAD)*, 2004, pp. 80–87.

[3] A. J. Laub and W. F. Arnold, “Controllability and observability criteria for multivariable linear second-order models,” *IEEE Trans. Automat. Contr.*, vol. 29, pp. 163–165, 1987.

[4] D. G. Meyer and S. Srinivasan, “Balancing and model reduction for second-order form linear systems,” *IEEE Trans. Automat. Contr.*, vol. AC-41, pp. 1632–1644, 1996.

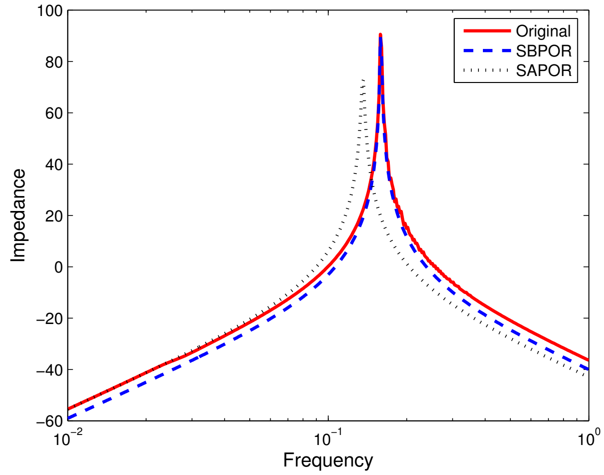

Figure 7: Comparison of frequency responses

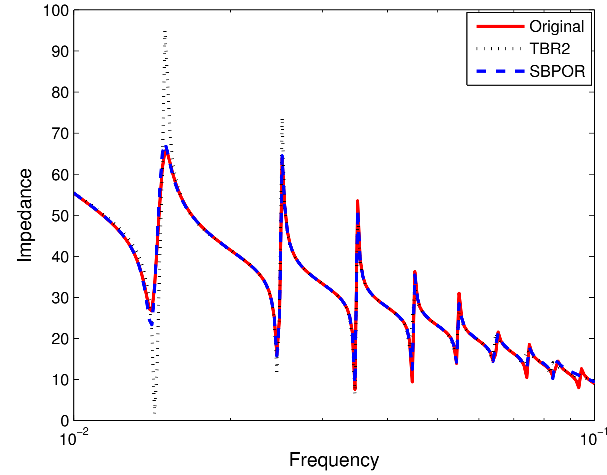

Figure 9: Comparison of frequency responses

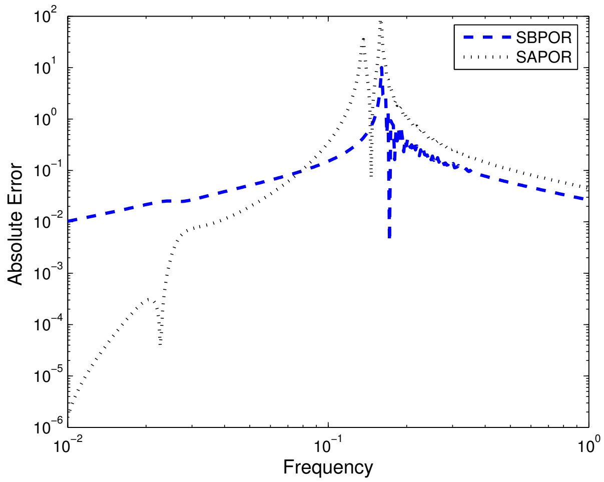

Figure 8: Comparison of absolute errors

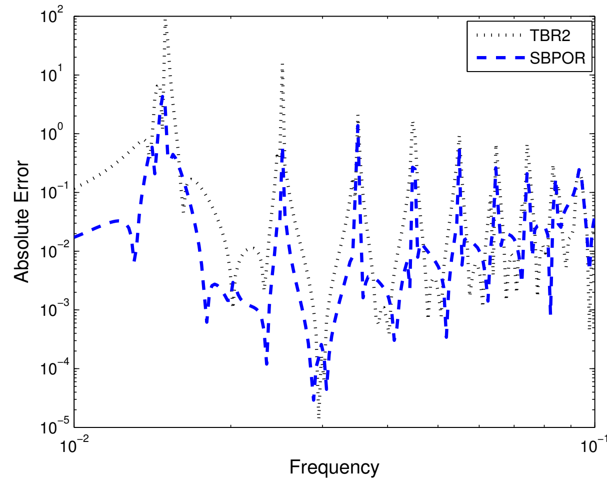

Figure 10: Comparison of absolute errors

[5] B. Moore, "Principle component analysis in linear systems: Controllability, and observability, and model reduction," *IEEE Trans. Automat. Contr.*, vol. 26, no. 1, pp. 17–32, 1981.

[6] A. Odabasioglu, M. Celik, and L. Pileggi, "PRIMA: Passive reduced-order interconnect macromodeling algorithm," *IEEE Trans. on Computer-Aided Design of Integrated Circuits and Systems*, pp. 645–654, 1998.

[7] J. R. Phillips, L. Daniel, and L. M. Silveira, "Guaranteed passive balanced transformation for model order reduction," *IEEE Trans. on Computer-Aided Design of Integrated Circuits and Systems*, vol. 22, no. 8, pp. 1027–1041, 2003.

[8] L. T. Pillage and R. A. Rohrer, "Asymptotic waveform evaluation for timing analysis," *IEEE Trans. on Computer-Aided Design of Integrated Circuits and Systems*, pp. 352–366, April 1990.

[9] B. N. Sheehan, "ENOR: model order reduction of RLC circuits using nodal equations for efficient factorization," in *Proc. Design Automation Conf. (DAC)*, 1999, pp. 17–21.

[10] M. Silveira, M. Kamon, I. Elfadel, and J. White, "A coordinate-transformed Arnoldi algorithm for generating guaranteed stable reduced-order models of RLC circuits," in *Proc. Int. Conf. on Computer Aided Design (ICCAD)*, 1996, pp. 288–294.

[11] T. Stykel, "Gramian-based model model reduction for descriptor systems," *Math. Control Signals Systems*, vol. 16, pp. 297–319, 2004.

[12] Y. Su, J. Wang, X. Zeng, Z. Bai, C. Chiang, and D. Zhou, "SAPOR: second-order arnoldi method for passive order reduction of RCS circuits," in *Proc. Int. Conf. on Computer Aided Design (ICCAD)*, 2004, pp. 74–79.

[13] N. Wang and V. Balakrishnan, "Fast balanced stochastic truncation via a duadratic extension of the alternating direction implicit iteration," in *Proc. Int. Conf. on Computer Aided Design (ICCAD)*, 2005, pp. 801–805.